Alex Dils

Berkeley, CA | dils@berkeley.edu | alex-dils.com | Scholar | LinkedIn | GitHub

GitHub Website

Education

University of California, Berkeley — B.A. in Computer Science; GPA: 3.7/4.0 May 2029

Relevant Coursework

CS 61A: abstraction, recursion, higher-order functions, interpreters/macros; **CS 61B**: data structures, algorithms, Java, software engineering; **CS 61C**: computer architecture, machine structures, memory/OS support; **CS 70**: proofs, discrete math, modular arithmetic/crypto, probability/Bayes; **Calculus I–III**: differential/integral calculus, techniques/applications, series, multivariable/vector calculus, Green/Gauss/Stokes.

Technical Skills

Python, PyTorch, TensorFlow, scikit-learn, OpenCV, SimpleITK, Core ML, ARKit, Metal, VLMs, multimodal learning, medical imaging, model evaluation, prompt engineering, data pipelines, FastAPI, Node/TypeScript, React, SQL/Postgres, Supabase/PostGIS, Docker, Slurm

Experience

Machine Learning Engineer | Logitech (ML@B) Feb 2026 – Present

- Built a multi-model **VLM evaluation harness** for meeting-room readiness with structured prompting/parsing, 28 ablations (7 models $\times$ 4 strategies), and **0% parse failures** across 504 predictions.
- Ran latency/accuracy benchmarking across VLMs and CV baselines (YOLOv11, MobileNet-SSD); found SmolVLM2-2.2B matched 4B–8B models while running **27$\times$ faster**, shaping edge-inference deployment tradeoffs.

Founder & CEO, Infrastruct — Agentic Workflow Automation Platform 2026 – Present

- Built an **AI agent platform** with Electron desktop observer + Node backend that mines repeated workflows, matches automation templates, and promotes high-confidence findings into approval-gated automations.
- Designed agent manifests, connector scopes, evidence packs, rollback/idempotency plans, always-on Codex/Telegram agent loops, tracing, OAuth/SSO, billing, deployment, and SOC 2 readiness workflows.

Canary CREST Research Scholar, Stanford Canary Center / Gevaert Lab — Stanford, CA Summer 2026

- Selected for NCI-funded 10-week cancer research program; applying deep learning, foundation models, and multimodal biomedical data fusion under Olivier Gevaert's lab.

CTO & Co-Founder, Appraise AI — Berkeley, CA | appraiseai.co Nov 2025 – Present

- Led ML roadmap, architecture, and enterprise technical strategy for a resale intelligence startup; raised **$15K** pre-seed.
- Architected **multimodal pricing engine** (computer vision + NLP) with **>95% top-3 accuracy**; built Scrapy/Playwright ingestion collecting **500K+ listings/day**.

Research Intern, Stanford Center for Biomedical Informatics Research — Remote Jun 2022 – Present

- Developed **bias-mitigation** and domain-robustness methods for skin-lesion segmentation; built finite-element lung-motion augmentations that **improved Dice by >0.05** on tumor segmentation.
- Contributed to Digital Twin research **featured on Stanford Medicine's homepage**; productionized medical-imaging workflows with PyTorch, TensorFlow, SimpleITK, Slurm, OpenCV, and scikit-learn.

Leadership

Machine Learning at Berkeley: Internal & Research Committees; **StreetCode Academy**: designed and taught Python curriculum to 30+ students across four 8-week cohorts.

Projects

SiteFit / RealSpace (Real Estate Intelligence & 3D Capture) 2026

- Built a CRE intelligence MVP with FastAPI scoring, Supabase/PostGIS, public-data ingestion, Stripe/CRM flows, **181 tests**, and a **50-location benchmark**; developed iOS ARKit/Core ML Depth Anything/Metal non-LiDAR room-mesh capture.

PianoKeys & Calshi | PianoKeys | calshi.app 2025–2026

- Built SwiftUI/Core ML on-device music transcription with Basic Pitch, MIDI/MusicXML export, and piano-roll playback; shipped Calshi as a React/Node/SQL full-stack product reaching **100+ active users**.

Selected Publications

- **Enhancing Medical AI with Physiologically-Informed Data Augmentation**, ICML 2026 submission (pending).
- **Eye For An Eye: A Deep-Learning and Analytical Method to Spatializing Stereoscopic Images**, National High School Journal of Science, 2025.
- **Microplastic Identification Using AI-Driven Image Segmentation and GAN-Generated Ecological Context**, arXiv, 2024.
- **Addressing Bias and Confounders in AI-based Image Diagnosis — A Study of 117,610 Skin Lesions**, Cell (in revision), 2025.
- **Integrating Mechanistic Knowledge into Deep Learning for Improved Cancer Detection**, FEniCS Conference Proceedings, 2024.